Section 8.1 Exercises

Directions: For exercises 1 - 4, solve each triangle. Where applicable, round to one decimal place.

1. $A = 43°, C = 69°, a = 20$

2. $A = 35°, C = 73°, c = 20$

3. $a = 4,\ A = 60°,\ B = 100°$

4. $b = 10,\ B = 95°, C = 30°$

Directions: For exercises 5 - 7, solve for the missing side for each oblique triangle. Round each answer to two decimal places.

5. Find side b when $A = 37°,\ B = 49°,\ c = 5.$

6. Find side a when $A = 132°, C = 23°, b = 10.$

7. Find side c when $B = 37°, C = 21°,\ b = 23.$

Directions: For exercises 8 - 17, determine whether there is no triangle, one triangle, or two triangles. Then solve each triangle, if possible. Round each answer to one decimal place.

8. $A = 119°, a = 14, b = 26$

9. $C = 113°, b = 10, c = 32$

10. $b = 3.5,\ c = 5.3,\ C = 80°$

11. $a = 12,\ c = 17,\ \alpha = 35°$

12. $a = 20.5,\ b = 35.0,\ \beta = 25°$

13. $a = 7,\ c = 9,\ \alpha = 43°$

14. $a = 7, b = 3, \beta = 24°$

15. $b = 13, c = 5, \gamma = 10°$

16. $a = 2.3, c = 1.8, \gamma = 28°$

17. $\beta = 119°, b = 8.2, a = 11.3$

Directions: For exercises 18 - 20, use the Law of Sines to solve, if possible, for the missing side or angle for each triangle or triangles in the ambiguous case. Round each answer to one decimal place.

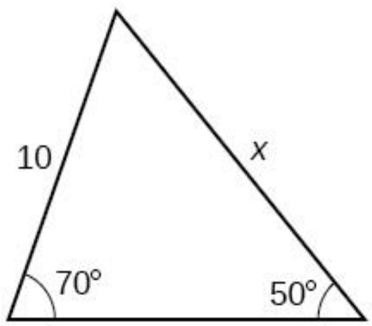
18. Find angle A when $a = 24, b = 5, B = 22°$.

19. Find angle A when $a = 13, b = 6, B = 20°$.

20. Find angle B when $A = 12°, a = 2, b = 9$.

Directions: For exercises 21 - 26, find the length of side x. Round the answer to one decimal place.

21.



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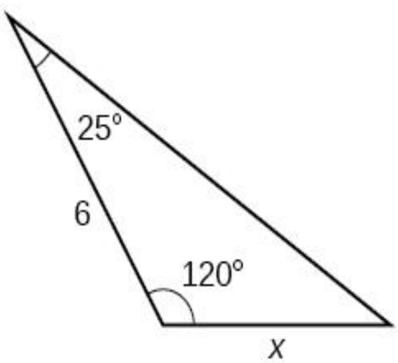

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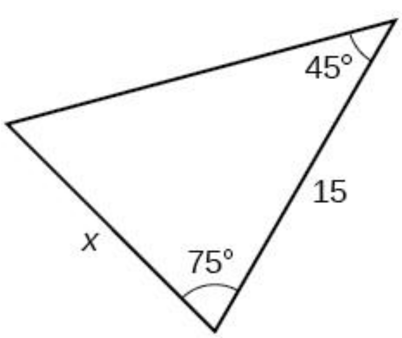

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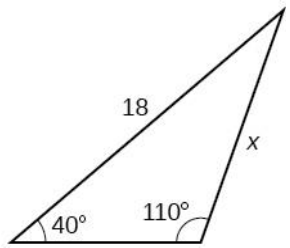

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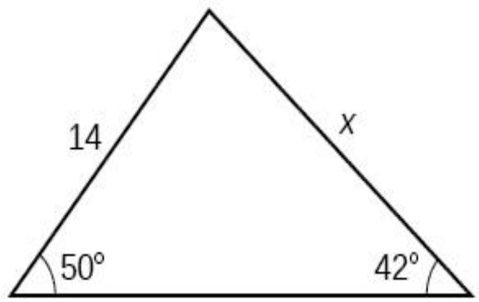

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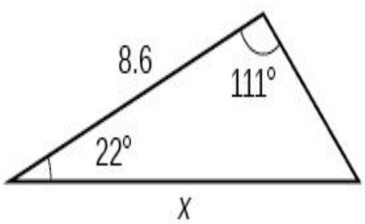
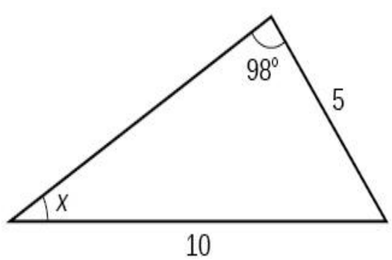


Directions: For exercises 27 - 32, find the measure of angle x, if possible. Round the answer to one decimal place.

27.



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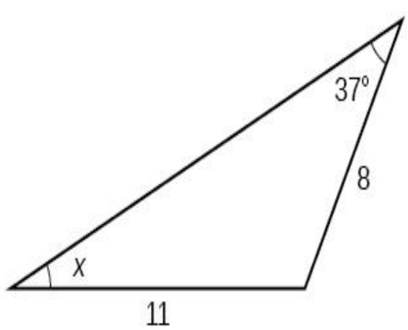

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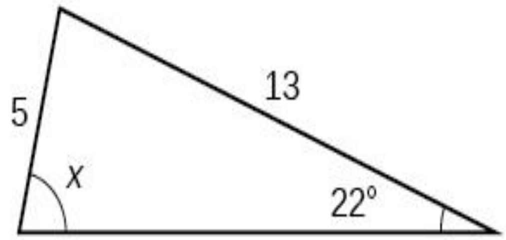

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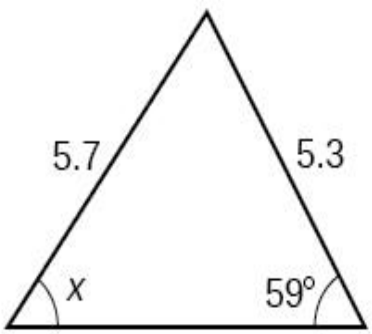

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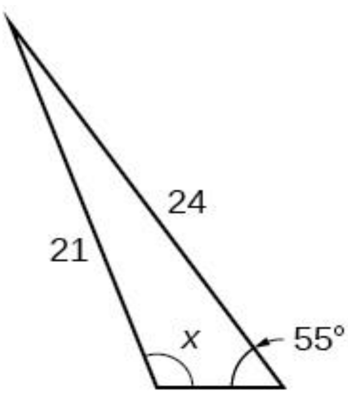

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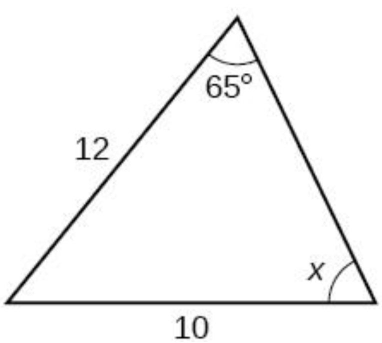

Directions: For exercise 33, solve the triangle. Round all answers to one decimal place.

33.